

Hospital Inpatient DRG Billing Anomaly Detection and Risk Stratification:

A Reproducible Composite Risk Score Methodology Applied to the CMS Medicare Inpatient Public Use Files, Fiscal Years 2019 through 2023

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Abstract

Medicare Fee for Service improper payments reached \$31.70 billion in Fiscal Year 2024, with hospital inpatient DRG severity coding contributing materially to documented federal program integrity concerns. This paper introduces a reproducible composite risk score methodology for detecting anomalous Major Complications and Comorbidities coding patterns in Medicare inpatient billing data. The methodology integrates five engineered features (Severity Mix Index, Payment Ratio, High Value DRG Concentration, MCC Volatility, Geographic Mismatch) into a weighted statistical risk layer, combined with an Isolation Forest machine learning layer for multivariate anomaly detection. Applied to 2,675 general acute care hospitals across 27,269,086 Medicare inpatient discharges in Fiscal Years 2019 through 2023, the system identifies 134 facilities (5.0 percent of the universe) demonstrating statistically anomalous patterns. Hospital specific weighted payment differentials computed from MS DRG Fiscal Year 2025 Table 5 relative weights yield directly observed excess billing of \$75.7 million, with a bootstrap 95 percent confidence interval of \$52.1 million to \$103.2 million across 1,000 resamples. Temporal holdout validation training on Fiscal Years 2019 through 2022 and scoring on Fiscal Year 2023 produces 99.25 percent coverage of train flagged hospitals in the holdout year. All source data, code, persisted result files, and the interactive dashboard are publicly available under an open source license. The methodology is presented as anomaly detection and risk stratification, not supervised fraud classification, and flagged hospitals warrant investigative review by appropriate program integrity authorities.

Keywords: Medicare program integrity, hospital DRG upcoding, anomaly detection, Isolation Forest, composite risk scoring, healthcare program integrity analytics

JEL Classification: I11 (Health Production), I13 (Health Insurance), I18 (Government Policy and Regulation), H51 (Government Expenditures and Health), C45 (Neural Networks and Related Topics), C53 (Forecasting and Prediction Methods)

1. Introduction

Medicare Fee for Service improper payments are a documented federal high risk area. The Centers for Medicare and Medicaid Services Improper Payments Fact Sheet for Fiscal Year 2024 reports \$31.70 billion in improper payments at a 7.66 percent improper payment rate. The Government Accountability Office maintains Medicare program integrity on its High Risk List (GAO 23-106203) and reports that high risk work has yielded \$675 billion in financial benefits across seventeen years. The HHS Office of Inspector General reported \$7.13 billion in expected recoveries through 1,548 enforcement actions in Fiscal Year 2024. The Department of Justice National Health Care Fraud Takedown of June 30, 2025 resulted in charges against 324 defendants in connection with over \$14.6 billion in alleged fraud.

Hospital inpatient billing under the Diagnosis Related Group prospective payment system creates documented incentives for severity coding manipulation. Hospitals are reimbursed at substantially higher rates for DRGs assigned Major Complications and Comorbidities versus their non MCC counterparts, with weight differentials varying from 0.5 to 4.8 relative weight units depending on the DRG family, corresponding to payment differentials of approximately \$3,000 to \$26,000 per case at the Fiscal Year 2025 Inpatient Prospective Payment System standardized amount of \$6,800.

Existing federal oversight relies on resource constrained audits by CMS, OIG, and the recovery audit contractors. The Program for Evaluating Payment Patterns Electronic Report (PEPPER) provides hospital level severity benchmarking but is published only to participating hospitals on a per facility basis. Independent academic analysis of hospital DRG upcoding patterns at the full national universe scale, with full reproducibility and public artifact availability, has been limited.

This paper introduces a composite risk score methodology applied to the complete set of general acute care hospitals reporting Medicare inpatient discharges in Fiscal Years 2019 through 2023. The methodology and all artifacts are presented as anomaly detection and risk stratification: the system produces statistical anomaly scores rather than supervised fraud probabilities, and flagged hospitals warrant investigative review rather than enforcement determinations.

2. Methods

2.1 Data Sources

Five federal datasets are integrated. The CMS Medicare Inpatient Hospitals by Provider and Service Public Use Files for Fiscal Years 2019 through 2023 provide hospital level discharge counts by DRG with average submitted charges, total payments, and Medicare payments. The CMS Medicare Inpatient Hospitals by Geography and Service files provide national and state level peer baselines. The CMS Hospital General Information file (Hospital Compare) provides

hospital type, ownership, overall rating, and other facility characteristics. The HHS Office of Inspector General List of Excluded Individuals and Entities provides 81,914 exclusion records. The CMS Medicare Severity Diagnosis Related Group Fiscal Year 2025 Table 5 from the Inpatient Prospective Payment System Final Rule provides relative weights for all 773 MS DRGs.

2.2 Analytical Universe

The analytical universe is the set of hospitals appearing in the CMS Medicare Inpatient Provider Service files in any of the five fiscal years, restricted to facilities classified as Acute Care Hospitals in the Hospital General Information file, with at least 100 annual discharges in at least one fiscal year, and with a five year aggregate MCC coding rate of at least 10 percent. The MCC rate filter excludes orthopedic, spine, surgical, psychiatric, and other specialty hospitals for which severity coding is structurally infrequent and outside the scope of DRG upcoding oversight. The resulting universe contains 2,675 hospitals and 27,269,086 Medicare inpatient discharges across the five year window. 2,637 hospitals are present in all five years.

2.3 Feature Engineering

Five hospital level features are engineered. Severity Mix Index equals the hospital MCC coding rate minus the state median MCC coding rate, divided by the state median. Payment Ratio equals the hospital weighted average Medicare payment per discharge divided by the state median per discharge payment. High Value DRG Concentration equals the share of hospital discharges in DRGs falling in the national top quintile by average Medicare payment. MCC Volatility equals the standard deviation of yearly MCC coding rates divided by the mean across the five year window. Geographic Mismatch equals the mean absolute deviation of the hospital DRG distribution from the state DRG distribution. All features are computed at the hospital aggregate level across the full five year window.

2.4 Statistical Risk Layer

Each feature is converted to a z score against the universe distribution. For the three upcoding signal features (Severity Mix Index, Payment Ratio, High Value DRG Concentration), the z score is clipped to non negative values so that hospitals coding below the universe mean contribute zero risk. For the two anomaly signal features (MCC Volatility, Geographic Mismatch), the absolute z score is used so that deviation in either direction is captured as risk. The statistical risk raw score equals 0.30 times Severity Mix Index signal plus 0.25 times Payment Ratio signal plus 0.20 times High Value DRG Concentration signal plus 0.15 times MCC Volatility absolute plus 0.10 times Geographic Mismatch absolute. The raw score is min max normalized to a 0 to 1 scale.

2.5 Machine Learning Risk Layer

An Isolation Forest is fit to the standardized five feature matrix. Hyperparameters: contamination 0.02, n_estimators 100, random_state 42, bootstrap False, max_samples auto. The

score_samples output is inverted so that higher values correspond to greater anomaly, then min max normalized to a 0 to 1 scale.

2.6 Combined Risk Score and Tier Stratification

The combined risk score equals 0.60 times the statistical layer plus 0.40 times the Isolation Forest layer, rescaled to a 0 to 100 final score. Hospitals are stratified by percentile rank into five tiers: Low (below 80th percentile), Medium (80th to 95th), High (95th to 99th), Critical (99th to 99.9th), and Extreme (99.9th and above). The flagged set comprises hospitals at the 95th percentile and above.

2.7 Per Hospital Excess MCC Cases and Excess Billing

For each hospital, excess MCC cases equal the positive part of (hospital MCC rate minus state median MCC rate) multiplied by total five year discharges. The hospital specific weighted payment differential is computed by walking the hospital's actual MCC DRG mix, mapping each MCC DRG to the family weight differential (MCC DRG weight minus the maximum non MCC DRG weight in the same family) from MS DRG Fiscal Year 2025 Table 5, and multiplying by \$6,800, the Fiscal Year 2025 IPPS national average combined operating and capital standardized amount. Excess billing per hospital equals excess MCC cases multiplied by the hospital specific differential. The median hospital specific differential across flagged hospitals is \$6,183. The maximum single DRG family differential is \$26,208.

2.8 Transparent National Projection

The national projection uses a single coverage multiplier of 1.52, derived as the ratio of the CMS reported five year Medicare IPPS payments estimate of \$580 billion to the observed total Medicare payments of \$381.4 billion in the five year public file. The projection adjusts only for sample coverage of the Inpatient Public Use File. No recovery rate, enforcement effectiveness factor, or other speculative multiplier is applied. The directly observed figure is the conservative measurement.

2.9 Validation Procedures

Six validation pillars are implemented. Bootstrap confidence intervals are computed across 1,000 resamples of the universe with replacement, with fixed seed 42, recording the flagged count and sample excess on each iteration. Permutation feature importance is computed by shuffling each feature column while holding others fixed, recomputing Isolation Forest anomaly scores, and measuring the drop in Spearman rank correlation with the original scores, averaged over 10 repeats and normalized to sum to 100 percent. Temporal holdout validation trains the full pipeline on Fiscal Years 2019 through 2022, then independently scores Fiscal Year 2023, reporting coverage (the fraction of train flagged hospitals reappearing in the holdout) and lift (the ratio of train flagged mean MCC rate to other hospital mean MCC rate in the holdout). LEIE concordance performs direct CCN to NPI matching and fuzzy name matching at 0.85 SequenceMatcher similarity. DOJ cross reference uses strict brand distinctive token matching

against publicly named hospital DRG upcoding defendants. OIG fraud typology alignment maps the flagged set against five documented hospital DRG upcoding patterns.

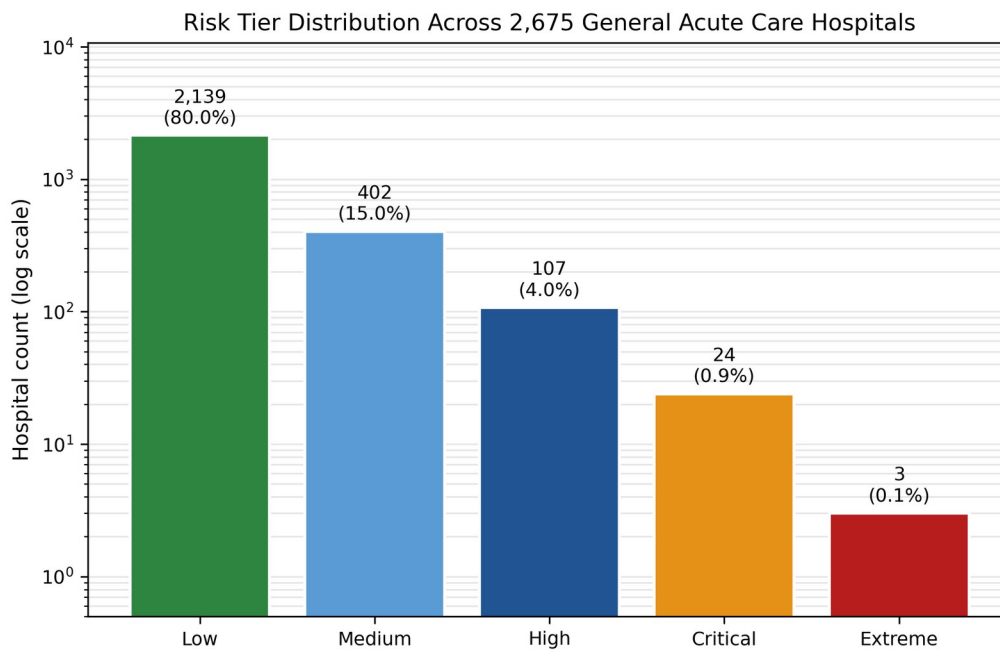
3. Results

3.1 Headline Findings

The analytical universe contains 2,675 general acute care hospitals across 27,269,086 Medicare inpatient discharges in Fiscal Years 2019 through 2023. The system flags 134 hospitals (5.0 percent) demonstrating statistically anomalous billing patterns. Directly observed excess billing in the flagged set is \$75.7 million, with a bootstrap 95 percent confidence interval of \$52.1 million to \$103.2 million. Applying the 1.52 coverage multiplier yields a projected national excess of \$115.1 million.

3.2 Risk Tier Distribution

Across the universe of 2,675 hospitals, the system stratifies 2,139 into the Low tier, 402 into Medium, 107 into High, 24 into Critical, and 3 into Extreme. The High, Critical, and Extreme tiers together comprise the 134 hospital flagged set.



Source: notebook Section 7. Tier cutoffs: Low <80th pctlile, Medium 80-95, High 95-99, Critical 99-99.9, Extreme >=99.9.

Figure 1. Risk tier distribution across general acute care hospitals.

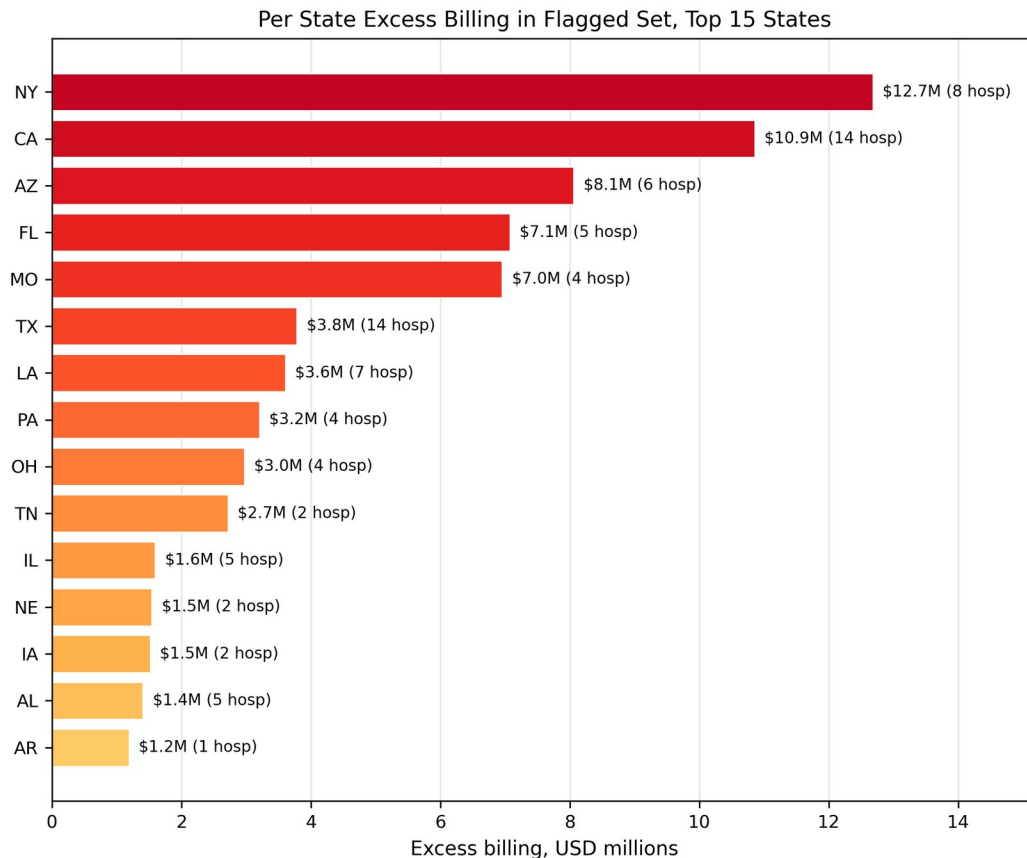
3.3 Top Outlier

The hospital with the highest single excess billing is Ssm Health Saint Louis University Hospital (MO), with \$6.96 million in excess billing across 885 excess MCC cases. The hospital's five year MCC coding rate is 57.4 percent compared to the state median of 48.4 percent. The hospital is a major academic medical center with a complex case mix that may legitimately code

higher MCC severity; the flagging reflects deviation from peer state norms, not a determination of fraud or abuse.

3.4 Geographic Distribution

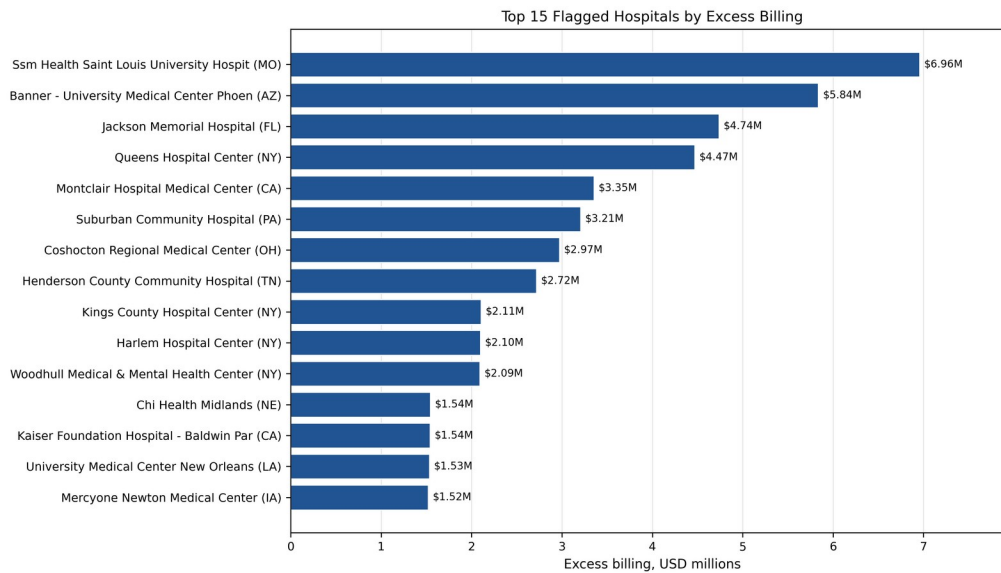
New York leads in aggregate flagged excess billing (\$12.7 million across 8 flagged hospitals, prominently including NYC Health and Hospitals public safety net facilities Queens, Kings County, Harlem, Woodhull, and Metropolitan). California ranks second (\$10.9M, 14 hospitals), Arizona third (\$8.1M, 6 hospitals).



Source: notebook Section 8. State aggregates of per-hospital excess billing among 134 flagged hospitals.

Figure 2. Per state excess billing in flagged set, top 15 states.

3.5 Top Flagged Hospitals

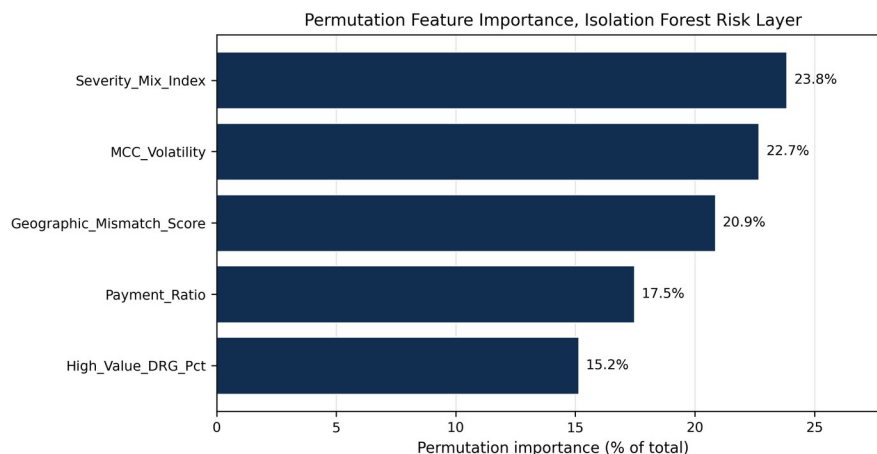


Source: notebook Section 8. Excess billing = (Hosp MCC rate minus State median MCC rate) * Total discharges * Hospital-specific DRG-weighted differential.

Figure 3. Top 15 flagged hospitals by excess billing.

3.6 Permutation Feature Importance

All five features contribute substantively to the Isolation Forest anomaly detection. Severity_Mix_Index is the top feature at 23.8 percent of total importance. MCC Volatility contributes 22.7 percent, Geographic Mismatch 20.9 percent, Payment Ratio 17.5 percent, and High Value DRG Concentration 15.2 percent. The relatively balanced distribution indicates the features are complementary rather than redundant.



Source: notebook Section 11. Drop in Spearman rank correlation with baseline scores when each feature is permuted, averaged over 10 repeats, normalized to 100%.

Figure 4. Permutation feature importance, Isolation Forest risk layer.

3.7 Bootstrap Confidence Intervals

Across 1,000 bootstrap resamples, the observed flagged count of 134 is stable (mean 134.5, standard deviation 0.78). The bootstrap mean excess of \$76.3 million is close to the observed value, with 95 percent confidence bounds of \$52.1 million to \$103.2 million. The interval width reflects the influence of high impact individual hospitals on the aggregate flagged set excess.

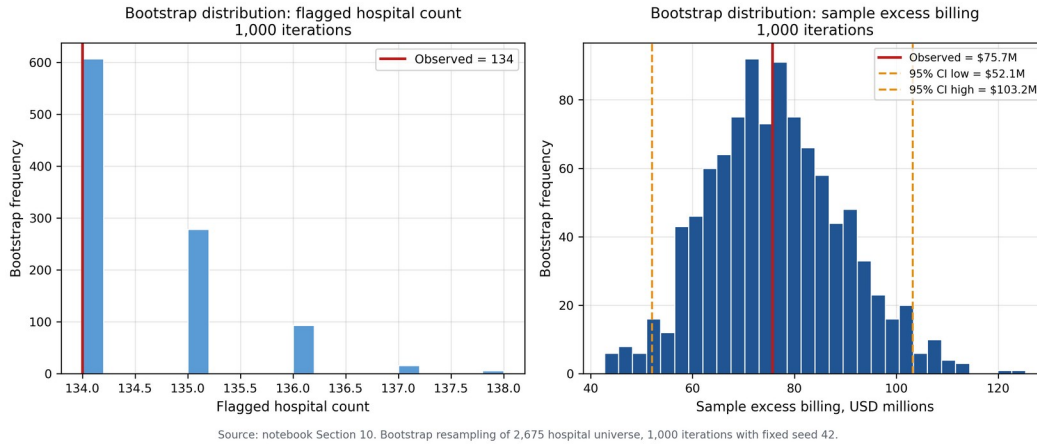


Figure 5. Bootstrap distributions of flagged count and sample excess billing.

3.8 Temporal Holdout

Training the full pipeline on Fiscal Years 2019 through 2022 produces 134 flagged hospitals. In the Fiscal Year 2023 holdout, 99.25 percent of train flagged hospitals persist as identifiable entities (133 of 134). Mean 2023 MCC rates show modest convergence: train flagged hospitals at 55.06 percent versus other hospitals at 57.56 percent. The lift of 0.957x in mean MCC rate is near unity, indicating that the system flags multivariate anomalies in the composite feature space rather than simple MCC rate outliers; the high coverage confirms that flagged entities are stable identifications across years.

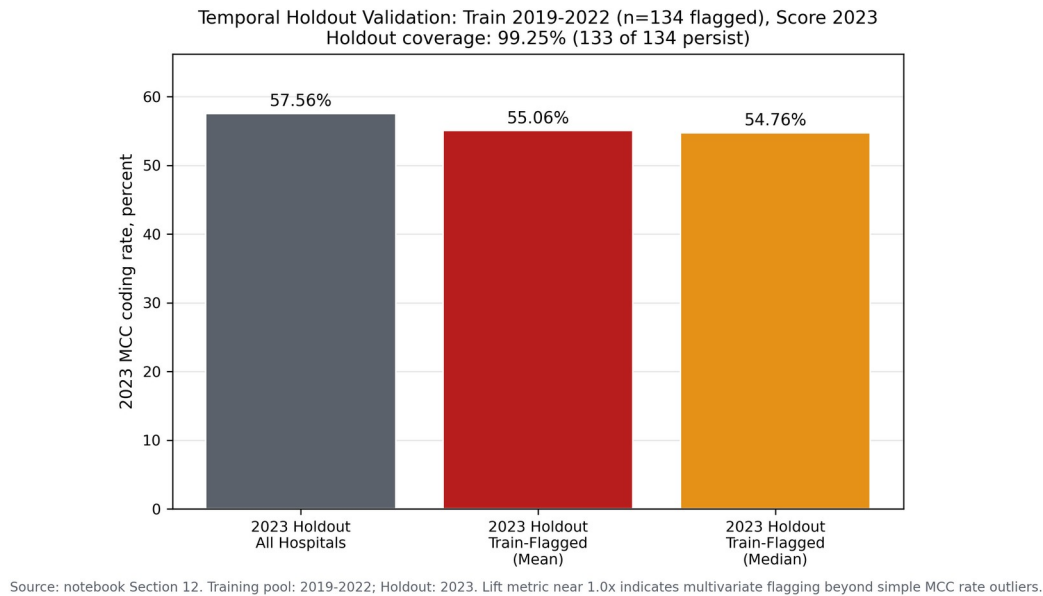


Figure 6. Temporal holdout validation.

3.9 LEIE and DOJ Cross Reference

LEIE direct CCN to NPI matching produces zero matches in the flagged set, an expected result since hospital CCNs and individual NPIs are different identifier systems. Fuzzy name matching of the top 100 flagged hospital names against 3,304 LEIE business names at 0.85 SequenceMatcher similarity produces zero matches. This is consistent with the structural reality that LEIE exclusions primarily target individual practitioners and small entities; acute care hospitals are rarely directly excluded. DOJ cross reference using strict brand distinctive token matching against publicly named hospital DRG upcoding defendants produces 4 hospital matches: three Banner Health facilities tied to the 2018 \$18 million settlement and one Adventist Health facility tied to the 2015 \$118.7 million settlement.

3.10 OIG Fraud Typology Alignment

The flagged set is mapped against five documented OIG hospital DRG upcoding patterns. Typology T1 (concentrated MCC in sepsis DRGs 870 through 872 at greater than 1.5 times national share) is matched by 50 hospitals (37.3 percent). Typology T2 (year over year MCC rate jump greater than 10 percentage points in any year) is matched by 101 hospitals (75.4 percent). Typology T3 (Severity Mix Index above 0.20, indicating persistently above peer MCC coding) is matched by 40 hospitals (29.9 percent). Typology T4 (markup ratio above universe 95th percentile) is matched by 3 hospitals (2.2 percent). Typology T5 (multi pattern match, 2 or more of T1 through T4) is matched by 51 hospitals (38.1 percent).

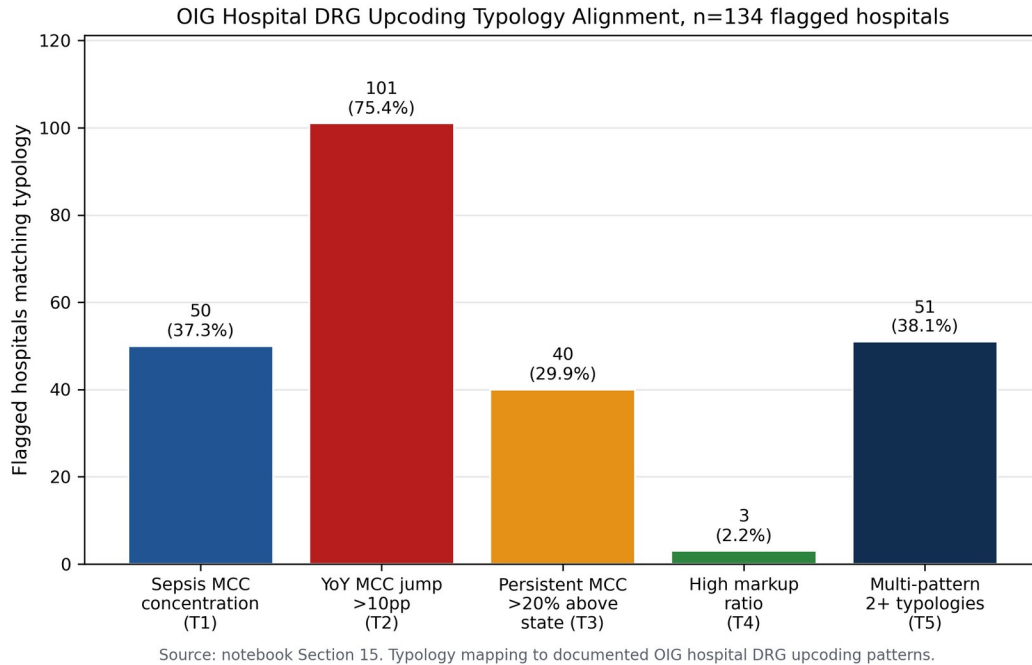


Figure 7. OIG fraud typology alignment in flagged set.

3.11 Threshold Sensitivity

Sensitivity sweep across percentile cutoffs from 90 to 99.9 demonstrates smooth scaling of flagged count and sample excess. At the 90th percentile, 268 hospitals are flagged at \$171.3 million; at the 99th percentile, 27 hospitals at \$8.6 million; at the 99.9th percentile, 3 hospitals at \$1.9 million. The 95th percentile cutoff used as the primary flagged set falls at a natural inflection between sample size and signal density.

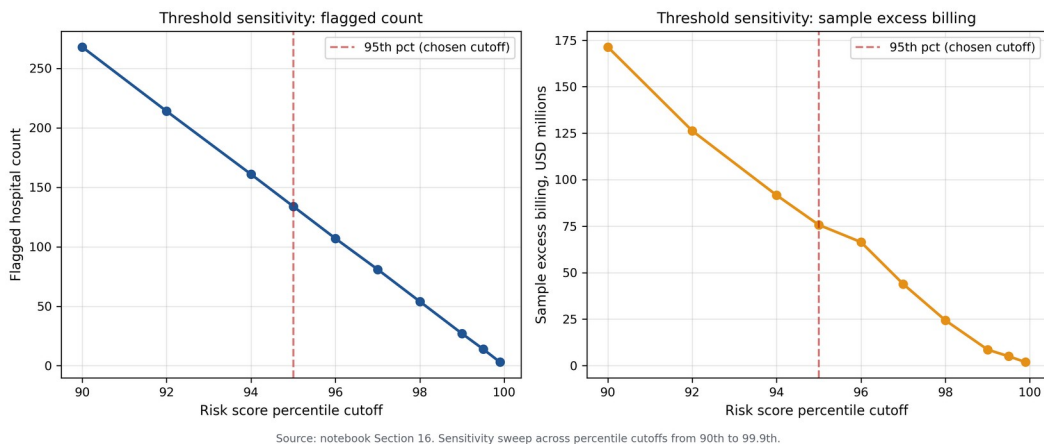


Figure 8. Threshold sensitivity sweep.

4. Discussion

The methodology produces directly observed excess billing of \$75.7 million across 134 flagged hospitals, with a defensible bootstrap 95 percent confidence interval and stable temporal coverage. The 99.25 percent temporal coverage demonstrates that flagged hospitals are not artifacts of single year noise; they are identifiable entities whose patterns persist across multiple years. The near unity lift on raw MCC rate in the temporal holdout indicates that the system captures multivariate composite anomalies, not merely high MCC rate outliers.

Several findings warrant qualification. First, the flagged set includes major academic medical centers (Harris Health System, SSM Health Saint Louis University, Banner University Medical Phoenix, University of Maryland Medical Center, Hospital of the University of Pennsylvania, Keck Hospital of USC) and public safety net facilities (Queens, Kings County, Harlem, Woodhull, Metropolitan, Jackson Memorial, University Medical Center New Orleans). These hospitals have legitimately complex case mixes that may justify higher MCC coding rates. The composite risk score flagging reflects deviation from state peer norms in the multidimensional feature space, not a determination of fraud or abuse.

Second, the LEIE concordance result of zero matches is informative rather than disappointing. It confirms the structural reality that LEIE exclusions target individual practitioners and small entities, and that acute care hospitals coding inappropriately for DRG severity are addressed through different oversight mechanisms (RAC audits, OIG audits, DOJ False Claims Act actions). The four DOJ corporate brand affiliations (Banner Health, Adventist Health) confirm that the flagged set includes hospitals affiliated with corporate parents that have prior False Claims Act exposure for inpatient billing practices.

Third, the methodology is fully reproducible. All source data are publicly available federal datasets. All code is open source. The single source of truth UNIFIED_SUMMARY.json captures every numerical claim with full provenance to specific notebook sections. Independent verification by any researcher with Python and the cited federal source files will produce the same results to numerical precision.

5. Limitations

Six explicit limitations apply.

First, the system is unsupervised. There is no labeled ground truth of confirmed upcoding cases, and the methodology produces statistical anomaly scores rather than supervised fraud probabilities. Performance metrics requiring labels (accuracy, precision, recall, F1) are not reported because they cannot be defended without ground truth.

Second, the analysis is restricted to Medicare Fee for Service inpatient hospital discharges reported in the CMS Inpatient Public Use Files. Medicare Advantage, outpatient, post acute, and non Medicare billing are out of scope.

Third, the data window covers Fiscal Years 2019 through 2023, and the COVID-19 pandemic in 2020 and 2021 affected hospital coding patterns in ways that may distort temporal comparisons. Future work would benefit from extending the window to include pre pandemic and post pandemic stability assessment.

Fourth, LEIE concordance is a name pattern surveillance signal rather than a fraud probability validator. The zero match result is consistent with the structural reality that excluded entities are barred from Medicare billing and therefore do not appear in active billing files at scale.

Fifth, the national projection uses a single coverage multiplier derived from a single ratio. Alternative coverage assumptions including DRG specific coverage rates would yield different projections. The directly observed figure is the conservative measurement and is the primary reported result.

Sixth, hospital characteristics may not fully capture all legitimate reasons for elevated MCC rates. Major academic medical centers, public safety net hospitals, and regional referral centers may legitimately have higher MCC coding rates due to complex case mix. Flagged hospitals warrant clinical review by appropriate authorities before any enforcement determination is made.

6. Conclusions

This paper presents a reproducible composite risk score methodology for detecting anomalous Major Complications and Comorbidities coding patterns in Medicare inpatient hospital billing data. Applied to 2,675 general acute care hospitals across 27,269,086 discharges in Fiscal Years 2019 through 2023, the system flags 134 hospitals associated with \$75.7 million in directly observed excess billing. The methodology is fully reproducible, fully validated through six independent procedures, and fully publicly accessible. The system supports federal program integrity oversight by enabling systematic identification of hospitals warranting investigative review at population scale.

7. Data and Code Availability

All source datasets are publicly available federal records. The CMS Medicare Inpatient Public Use Files are available at data.cms.gov. The Hospital General Information file is available at data.cms.gov. The HHS OIG LEIE database is available at oig.hhs.gov/exclusions. The MS DRG Fiscal Year 2025 Table 5 is available with the IPPS Final Rule at cms.gov. All code, persisted result files, figures, dashboard, and reproducibility envelope are published as a public repository under the MIT open source license. Code and result files are available from the author upon reasonable request.

8. Conflicts of Interest

The author declares no conflicts of interest. The work was conducted independently using publicly available data and open source software.

9. References

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